

Image Segmentation Algorithm

K-Means · Fuzzy C-Means · RFCM + GLCM Feature Extraction

Based on: Maji & Pal — Maximum Class Separability for Rough-Fuzzy C-Means Based Brain MR Image Segmentation

Step	Phase	Output
1	Feature Extraction (13 features/pixel)	Feature matrix (H×W, 13)
2	Discriminant Analysis Initialization	Initial centroids $v_1 \dots v_c$
3A	K-Means (Hard C-Means)	Hard label map + DB, Dunn, β
3B	Fuzzy C-Means	Soft membership map + indices
4	Validity Evaluation	DB Index, Dunn Index, β Index

Phase 1 — Feature Extraction (13 Features per Pixel)

For every pixel $I(m,n)$ in the image, compute the following 13 features. Each pixel becomes an object x_j in R^{13} . The full dataset is $X = \{x_1, \dots, x_n\}$.

Feature 1 — Gray Value

The raw intensity value of the pixel. No computation needed.

Feature 2 — Homogeneity (First-Order, 3×3 Neighborhood)

$$H = 1 - (1 / 6 * (I_{\max} - I_{\min})) * \{ |I(m-1, n-1) + I(m+1, n+1) - I(m-1, n+1) - I(m+1, n-1)| + |I(m-1, n-1) + 2*I(m, n-1) + I(m+1, n-1) - I(m-1, n+1) - 2*I(m, n+1) - I(m+1, n+1)| \}$$

$I_{\max}$ and $I_{\min}$ are the global max and min gray values. High H = uniform region (entirely within one organ). Low H = boundary between regions.

Feature 3 — Edge Value (First-Order)

$$\Delta = (1/4) * \max \{ |I(m-1, n) + I(m-1, n+1) + I(m, n) + I(m, n+1) - I(m+1, n) - I(m+1, n+1) - I(m+2, n) - I(m+2, n+1)|, \\ |I(m, n-1) + I(m, n) + I(m+1, n-1) + I(m+1, n) - I(m, n+1) - I(m, n+2) - I(m+1, n+1) - I(m+1, n+2)| \}$$

Maximum of differences between average gray levels in horizontally and vertically adjacent 2×2 neighborhoods. High Δ = edge pixel. Low Δ = interior pixel.

Features 4–13 — Haralick Texture Features (Second-Order)

All 10 remaining features are derived from the Gray Level Co-occurrence Matrix (GLCM). The GLCM is built for each pixel using a 3×3 sliding window centered at that pixel.

Step A — Build the GLCM

Compute one GLCM per angular direction, then average them:

Direction	Row offset	Col offset	Neighbor
0°	0	+1	Right
45°	-1	+1	Top-right
90°	-1	0	Top
135°	-1	-1	Top-left

For each direction d : For every pixel (r,c) in the 3×3 window: a = gray value of pixel (r,c) b = gray value of neighbor in direction d $GLCM_d[a][b] += 1$ $GLCM_d[b][a] += 1$ # enforce symmetry Normalize $GLCM_d$ by dividing by its total sum $GLCM_{\text{final}} = (GLCM_0 + GLCM_{45} + GLCM_{90} + GLCM_{135}) / 4$ Size: $N_g \times N_g$ where N_g = number of gray levels (e.g. 256 for 8-bit)

Note: $P(i,j)$ means gray level i and j — NOT pixel positions. i and j are intensity values. The matrix size is $Ng \times Ng$.

Step B — Compute Marginal Distributions

```
px(i) = sum over j of P(i,j) # row marginal
py(j) = sum over i of P(i,j) # col
marginal
mu_x = sum_i [ i * px(i) ] # mean of px
mu_y = sum_j [ j * py(j) ] #
mean of py
sig_x2 = sum_i [ (i - mu_x)^2 * px(i) ]
sig_y2 = sum_j [ (j -
mu_y)^2 * py(j) ]
p_xpy(k) = sum_{i+j=k} P(i,j) for k = 2..2*Ng # sum dist
p_xmy(k) = sum_{|i-j|=k} P(i,j) for k = 0..Ng-1 # diff dist
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Phase 1 (cont.) — The 10 Haralick Features

Feature 4 — Angular Second Moment (ASM / Energy)

$$f1 = \sum_i \sum_j P(i,j)^2$$

Measures uniformity. High in smooth/uniform regions.

Feature 5 — Contrast

$$f2 = \sum_{n=0}^{Ng-1} n^2 * [\sum_{|i-j|=n} P(i,j)]$$

Local intensity variation. High when neighbors differ greatly.

Feature 6 — Correlation

$$f3 = [\sum_i \sum_j (i*j)*P(i,j) - \mu_x * \mu_y] / (\sigma_x * \sigma_y)$$

Linear dependency of gray levels. Ranges from -1 to +1.

Feature 7 — Inverse Difference Moment (IDM)

$$f4 = \sum_i \sum_j P(i,j) / (1 + (i-j)^2)$$

High when similar gray levels are close. Opposite of contrast.

Feature 8 — Sum Average

$$f5 = \sum_{k=2}^{2*Ng} k * p_{xpy}(k)$$

Mean of the sum distribution.

Feature 9 — Sum Variance

$$f6 = \sum_{k=2}^{2*Ng} (k - f5)^2 * p_{xpy}(k)$$

Variance of the sum distribution.

Feature 10 — Sum Entropy

$$f7 = - \sum_{k=2}^{2*Ng} p_{xpy}(k) * \log(p_{xpy}(k) + \text{eps})$$

Entropy of the sum distribution. eps = 1e-10 to avoid log(0).

Feature 11 — Second-Order Entropy

$$f8 = - \sum_i \sum_j P(i,j) * \log(P(i,j) + \text{eps})$$

Overall randomness of the GLCM. High = complex/random texture.

Feature 12 — Difference Variance

$$\mu_{xmy} = \sum_k [k * p_{xmy}(k)] \quad f_9 = \sum_{k=0}^{Ng-1} (k - \mu_{xmy})^2 * p_{xmy}(k)$$

Variance of the difference distribution.

Feature 13 — Difference Entropy

$$f_{10} = - \sum_{k=0}^{Ng-1} p_{xmy}(k) * \log(p_{xmy}(k) + \text{eps})$$

Entropy of the difference distribution.

#	Feature	Intuition	High value means
4	ASM	Uniformity	Smooth region
5	Contrast	Local variation	Strong edges
6	Correlation	Linear dependency	Structured texture
7	IDM	Local homogeneity	Uniform patches
8	Sum Average	Mean pair intensity	Bright region
9	Sum Variance	Texture coarseness	Coarse texture
10	Sum Entropy	Disorder in sums	Complex texture
11	Entropy	Overall disorder	Random texture
12	Diff Variance	Edge strength	Sharp boundaries
13	Diff Entropy	Boundary complexity	Irregular edges

Phase 2 — Discriminant Analysis Initialization

Replaces random centroid selection. Uses class separability maximization (Otsu's method) across all 13 features to find smart initial centroids, avoiding local minima. $L = 101$ discrete bins.

Step 1. Discretize each feature f_m into $L=101$ bins:

$$\text{Disc}(f_{mj}) = (L-1) * (f_{mj} - f_{m_min}) / (f_{m_max} - f_{m_min})$$

Step 2. For each feature f_m , compute histogram $h(z)$ = fraction of pixels with discretized value z , for $0 \leq z < L$.

Step 3. Find threshold T_m that maximizes $J(T)$:

Step 4. Binarize each feature using its threshold T_m :

$$f_{\text{bar_mj}} = 1 \text{ if } \text{Disc}(f_{mj}) \geq T_m \text{ } 0 \text{ otherwise}$$

Step 5. Repeat Steps 1-4 for all 13 features. Each pixel becomes a binary vector in $\{0,1\}^{13}$.

Step 6. For each unique binary pattern x_i , compute: $N(x_i)$ = count of pixels matching x_i $v_{\text{bar}}(x_i)$ = mean of original real-valued pixels sharing that pattern

Step 7. Sort unique patterns by $N(x_i)$ descending. Remove duplicates.

Step 8. Compute heuristic threshold T_r : $R = \sum_{i=1}^n 1/(N(x_i) - N(x_{i+1}))$ $T_r = R / \epsilon_{\text{tilde}}$ ($\epsilon_{\text{tilde}} = 0.5$) All patterns with $N(x_i) > T_r$ contribute $v_{\text{bar}}(x_i)$ as a candidate initial centroid.

Class Separability Formula $J(T)$:

$$J(T) = [P1(T) * P2(T) * (m1(T) - m2(T))^2] / [P1(T)*\text{sig1}^2(T) + P2(T)*\text{sig2}^2(T)]$$
$$P1(T) = \sum_{z=0}^T h(z) \quad P2(T) = 1 - P1(T) \quad m1(T) = (1/P1) * \sum_{z=0}^T z * h(z) \quad m2(T) = (1/P2) * \sum_{z=T+1}^{L-1} z * h(z)$$
$$\text{sig1}^2(T) = (1/P1) * \sum_{z=0}^T (z-m1)^2 * h(z) \quad \text{sig2}^2(T) = (1/P2) * \sum_{z=T+1}^{L-1} (z-m2)^2 * h(z)$$
$$T_m = \text{argmax}_T J(T)$$

Phase 3A — K-Means (Hard C-Means)

Uses the centroids $v_1 \dots v_c$ from Phase 2. Hard assignment — each pixel belongs to exactly one cluster.

Algorithm Steps:

Step 1: Initialize centroids $v_1 \dots v_c$ from Phase 2 output.

Step 2: For each pixel x_j , compute distance to every centroid: $d_{ij}^2 = ||x_j - v_i||^2$

Step 3: Assign pixel to nearest centroid: $\text{cluster}(x_j) = \text{argmin}_i d_{ij}$

Step 4: Update each centroid as mean of its assigned pixels: $v_i = (1/|C_i|) * \sum_{\{x_j \text{ in } C_i\}} x_j$

Step 5: Repeat Steps 2-4 until assignments no longer change.

Phase 3B — Fuzzy C-Means (FCM)

Soft assignment — each pixel has a membership degree μ_{ij} in $[0,1]$ for every cluster. Uses same initial centroids from Phase 2. Parameters: fuzzifier $m' = 2$, convergence threshold ϵ .

Algorithm Steps:

Step 1: Initialize centroids $v_1 \dots v_c$ from Phase 2. Set $m'=2$, ϵ , $t=1$.

Step 2: Compute memberships for all pixels x_j and clusters i : $d_{ij}^2 = ||x_j - v_i||^2$ $\mu_{ij} = [\sum_{k=1}^c (d_{ij}/d_{kj})^{2/(m'-1)}]^{(-1)}$ subject to: $\sum_i \mu_{ij} = 1$ for all j

Step 3: Update centroids: $v_i = [\sum_j (\mu_{ij})^{m'} * x_j] / [\sum_j (\mu_{ij})^{m'}]$

Step 4: Repeat Steps 2-3, incrementing t , until: $|\mu_{ij}(t) - \mu_{ij}(t-1)| \leq \epsilon$ for all i,j

Step 5: Final segmentation: assign pixel x_j to cluster with highest μ_{ij} .

Phase 4 — Cluster Validity Evaluation

Evaluate the quality of segmentation using three indices. Apply to both K-Means and FCM results. Paper uses $c=4$ as optimal for brain MRI, $w=0.95$, $m'=2.0$.

1. Davies-Bouldin Index (DB) — lower is better

$S(v_i)$ = mean intra-cluster distance for cluster i $d(v_i, v_k)$ = distance between centroids v_i and v_k
 $DB = (1/c) * \sum_{i=1}^c \max_{i \neq k} \{ (S(v_i) + S(v_k)) / d(v_i, v_k) \}$

Minimizes within-cluster distance and maximizes between-cluster separation.

2. Dunn Index — higher is better

$Dunn = \min_i \{ \min_{i \neq k} \{ d(v_i, v_k) / \max_l S(v_l) \} \}$

Maximizes inter-cluster separation relative to the largest cluster diameter.

3. Beta Index (β) — higher is better

v_bar = global mean of all n pixels $N = \sum_{i=1}^c \sum_{j=1}^{n_i} ||x_{ij} - v_bar||^2$ (total variation) $M = \sum_{i=1}^c \sum_{j=1}^{n_i} ||x_{ij} - v_i||^2$ (within-cluster variation) $\beta = N / M$

Ratio of total variation to within-cluster variation. Higher beta = more homogeneous clusters. Also increases with c .

Complete Algorithm Flow

IMAGE | v [Phase 1] Extract 13 features per pixel gray, homogeneity, edge, 10 Haralick (via GLCM) -> Feature matrix shape (H*W, 13) | v [Phase 2] Discriminant analysis initialization Discretize -> $h(z)$ -> maximize $J(T)$ -> binarize -> Count patterns -> threshold Tr -> Initial centroids $v_1...v_c$ | +-----+-----+ | v [Phase 3A] K-Means [Phase 3B] Fuzzy C-Means Hard label map Soft membership map | v [Phase 4] Evaluate: DB Index, Dunn Index, Beta Index Best result for brain MRI: $c=4$, $m'=2$, $w=0.95$

Paper parameters: $c=4$ clusters, $m'=2.0$ (fuzzifier), $w=0.95$ (lower bound weight), $L=101$ discrete bins, 13 features, ϵ =small convergence threshold.